

Alberto Blasco Ventas

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PROFESSIONAL SUMMARY

Generative AI Engineer with 2 years of experience building LLM-powered applications end-to-end—from RAG systems and prompt design/evaluation to secure API services. Proficient with LangChain and LangGraph for multi-step orchestration and agents; deploy vector search with Pinecone/Qdrant/Chroma and FastAPI backends containerized with Docker and automated via GitHub Actions. Experienced delivering React AI Agents and front-ends integrated with Azure Cognitive Services and SharePoint, shipping reliable, cost-aware solutions to production.

WORK EXPERIENCE

Cognizant — Generative AI Intern

May 2024 – Sep 2024 · Spain

- **Collaborated with the AI Engineering team** to design and implement Machine Learning and Generative AI solutions for document processing and automation.
- **Developed OCR-based pipelines using Azure Cognitive Services** and custom ML preprocessing to extract structured data from unstructured documents.
- **Performed data analysis, feature engineering, and applied supervised/unsupervised learning methods** with Pandas and NumPy to improve downstream model performance
- **Created and delivered technical presentations and reports** on Machine Learning and Generative AI topics, communicating results and insights to cross-functional teams and stakeholders.

Cognizant — Generative AI Developer

Sep 2024 – Present · Spain

- **Engineered end-to-end Generative AI solutions for Repsol**, building RAG pipelines, OCR systems, and intelligent AI agents integrated with enterprise workflows.
- **Utilized cutting-edge frameworks—LangChain, LangGraph, and CrewAI—to** design dynamic, multi-agent architectures leveraging Model Context Protocols (MCP) for secure tool and data access.
- **Built a document-processing assistant for Repsol using Azure OpenAI and Pinecone/Qdrant**, delivering grounded answers with citations and improved accuracy across internal datasets.
- **Developed modular FastAPI backends, containerized with Docker**, and automated deployments with GitHub Actions for reproducibility and scalability.
- **Collaborated with cross-functional teams at Repsol to integrate SharePoint for secure ingestion**, manage secrets via Azure Key Vault, and ensure robust, audit-compliant data pipelines.

PERSONAL PROJECTS

Research Project (Computer Vision)

- Performed feature engineering and fine-tuning of pre-trained models (YOLO-NAS, YOLOv9, CNNs, transformers) to evaluate and compare performance for drone detection in videos, optimizing the pipeline for accuracy (mAP).

OpenGenServer — FastAPI + Docker + HF Spaces ([HuggingFace](#))

- Backend service with JWT auth, endpoints for generation and embeddings, Dockerized, CI/CD with GitHub Actions, and deployment on Hugging Face Spaces.

Simple PDF RAG Chatbot — Gradio + Gemini + Qdrant ([HuggingFace](#))

- Ask questions over PDFs with citations and page references. Uses Gemini + text-embedding-004, local Qdrant, conversational memory and optional re-ranking.

EDUCATION

B.Sc. in Computer Science — Universidad Carlos III de Madrid (UC3M)

Academic Exchange: Åland University of Applied Sciences (Finland, 2022) | Stony Brook University (NY, USA, 2023)

SKILLS

Spanish (native) · English (advanced) Python · FastAPI · LangChain/LangGraph · CrewAI · RAG · Pinecone/Qdrant/Chroma
Azure OpenAI · Pandas/NumPy · AWS · LLM · AI Agents · MCP